Food Delivery Time Prediction

Machine Learning Classification Project Report

Project Type: Machine Learning / Data Science
Dataset: Food Delivery Time Prediction
Models Used: Gaussian Naive Bayes, K-Nearest Neighbors (KNN), Decision Tree

1. Introduction

Food delivery time can be affected by several factors such as distance, traffic, weather, order priority, vehicle type, and delivery conditions. In this project, machine learning is used to classify a delivery as either **Fast** or **Delayed**.

The main idea of the project is to take the information available for an order and use it to predict its delivery status. This gives a simple classification problem that can be used to understand how different machine learning algorithms behave on real-looking delivery data.

2. Objective

The objective of this project is to build and compare machine learning classification models for predicting whether a food delivery will be Fast or Delayed. The project also focuses on data preprocessing, feature preparation, model tuning, and evaluation using more than just accuracy.

3. Dataset Description

The dataset contains **200 orders and 15 original columns**. The columns include order details, customer and restaurant locations, distance, weather, traffic, delivery-person experience, ratings, delivery time, order cost, and tip amount.

Category	Examples
Order details	Order ID, Order Time, Order Priority
Location	Customer Location, Restaurant Location, Distance
Delivery conditions	Weather, Traffic, Vehicle Type
Ratings	Restaurant Rating, Customer Rating
Other numerical data	Delivery Person Experience, Order Cost, Tip Amount
Target-related column	Delivery Time

4. Data Preprocessing

Before training the models, the data was checked and prepared. Duplicate rows were removed. Missing numerical values were handled using the median, while missing categorical values were filled using the

mode.

A new target column called **Delivery_Status** was created. A delivery time greater than 30 minutes was treated as Delayed (1), while a delivery time of 30 minutes or less was treated as Fast (0).

The location columns contained latitude and longitude inside text values. These values were extracted separately for the customer and restaurant. Haversine distance was then calculated to demonstrate location-based feature engineering. The original Distance column was retained for modelling, while the calculated distance and raw location fields were removed before training.

Categorical variables were converted into numerical form using LabelEncoder. The data was then split into training and testing sets using an 80:20 split with stratification. StandardScaler was used for the models that required scaled features.

5. Class Distribution

After creating the target variable, the dataset contained **174 Delayed orders and 26 Fast orders**. This means that the two classes are not balanced. This imbalance is important when interpreting the model results because a model can get a relatively high accuracy by mostly predicting the larger class.

6. Machine Learning Models

Gaussian Naive Bayes: A simple probabilistic classification model. It was trained using the scaled features.

K-Nearest Neighbors: KNN classifies a sample based on nearby training samples. Different values of K were tested using 5-fold cross-validation. The selected value was **K = 7**.

Decision Tree: A tree-based model that makes decisions using feature values. Different maximum depths were tested using cross-validation. The selected depth was **2**, with min_samples_split set to 5.

7. Model Results

Model	Accuracy	Precision	Recall	F1 Score	AUC
Naive Bayes	87.5%	87.5%	100.0%	93.33%	0.3771
KNN	87.5%	87.5%	100.0%	93.33%	0.5314
Decision Tree	85.0%	87.18%	97.14%	91.89%	0.4743

8. Result Analysis

Naive Bayes and KNN both achieved 87.5% accuracy, while the Decision Tree achieved 85%. At first, these accuracy values may look quite good. However, the confusion matrices show that the models had difficulty recognizing the Fast class.

The test set contained 5 Fast deliveries and 35 Delayed deliveries. Naive Bayes and KNN classified all 40 test samples as Delayed. The Decision Tree also failed to correctly classify any of the 5 Fast deliveries. This shows why accuracy should not be considered by itself for this dataset.

The ROC-AUC values were 0.3771 for Naive Bayes, 0.5314 for KNN, and 0.4743 for Decision Tree. These values show that the models had limited ability to separate Fast and Delayed deliveries in the given test data.

9. Feature Importance

The Decision Tree feature-importance analysis showed that the trained tree mainly used **Customer Rating, Restaurant Rating, and Tip Amount** among the available features. This gives some idea of which variables were used more by the trained model, although feature importance does not by itself prove that these variables directly cause delivery delays.

10. Strengths and Limitations

Strengths: The project includes data cleaning, feature preparation, location-based feature engineering, three different classification models, cross-validation for model tuning, and several evaluation metrics. The Decision Tree also provides an easy way to inspect feature importance.

Limitations: The dataset is small and strongly imbalanced toward Delayed deliveries. The models therefore struggle to identify the smaller Fast class. The available features may also not contain enough information to separate the two classes reliably.

11. Possible Improvements

The project can be improved by collecting more Fast-delivery examples, applying suitable class-balancing methods, and adding more useful delivery-related information. Features related to detailed traffic conditions, weather, distance, delivery-person conditions, and other operational factors could be explored. Other classification algorithms could also be tested and compared.

12. Conclusion

This project used machine learning to predict whether a food delivery would be Fast or Delayed. The dataset was cleaned and prepared before applying Gaussian Naive Bayes, KNN, and Decision Tree models.

The results showed that the models achieved reasonably high accuracy, but they had difficulty identifying Fast deliveries. The main reason is the large difference between the number of Delayed and Fast orders in the dataset. The project therefore shows the importance of looking at confusion matrices, F1-score, and ROC-AUC instead of relying only on accuracy.

Overall, the project provides a basic machine learning approach for food delivery classification. With more balanced data and additional useful features, the prediction performance could be studied further.

Note: This report is based on the final submitted Jupyter Notebook and its recorded outputs.